THE FIRST RAILS

The first rail that was laid consisted of bars of flat iron, varying from ¼ to ¾ inch thickness, screwed upon longitudinal sleepers, which held in position by cross-bars or transoms. In 1850, however, 30,000 running feet of light rail bars, which had been selected in England by Major Baker, arrived at the works.

THE LOCOMOTIVE

On 22 December 1851, we started a locomotive, which I believe must have been the first engine of the sort that was ever used in India. The engine was in some way or other imperfect in the adjustment of its parts; it was continually getting out of order, and appeared to be another specimen of machinery for exportation to the colonies. The locomotive is small and compact; engine and tender in one frame; adapted from its water fuel arrangements to moderate distances only, and applicable to draw on a level line of rail, a train of ballast of wagons of an aggregate weight of 180 to 200 tonnes at a speed of four miles an hour. Of the parts most liable to fracture, we have duplicates, as well as of the wheels, both driving, tail, and leading. Its name is the ‘Thomason’. The executive engineer in charge of the Northern Division of the works, in writing to me on the subject of its applicability for the particular duties which the aqueduct works required from it, remarked: ‘The locomotive started first on 22 December 1851, at the Roorkee end of the railway. ’

THE WAGONS

A number of small wagons made up with side doors and without tilting apparatus arrived at Roorkee. These gave us the early means of commencing our earth-carrying operations. I had, however, brought with me from England tracings of what I thought to be the wagon best adapted to our works (with rear tilting), with the ironwork of one wagon complete.

FREEDOM FROM WHEELBARROWS AND BASKETS

The introduction of railroads and wagons relieved us from wheelbarrows and baskets, but for a long time kept us with men both for excavation and for propelling the wagons. As time advanced, horses in some measure took the place of men; and on 22 December 1851, we started a locomotive, which I believe to have been the first engine of the sort that was ever used in India. We, therefore, went through all the grades of carriage, with varied results, all of which, however, appeared to be dependent on local causes, or economical arrangements with regard to time.

Our methods of working, therefore, were:
First, with baskets and wheelbarrows
Second, with earth-wagons propelled by men
Third, with earth-wagons propelled by horses
Fourth, with earth-wagons propelled by a locomotive .